# RUSSELL JOSEPH MARTINEZ JR.

CURRICULUM VITAE

## Contact Information

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## Education

**Bachelor's Degree in Mathematics** Sep. 2021 – Present
*Advisors: Yi Ni* Caltech

GPA: 3.9/4.0. Relevant coursework: topological and smooth manifolds (Ma 109abc); sheaves and schemes (Ma 130a); homology theory, homotopy theory, and Riemannian geometry (Ma 151abc); Lie groups and their representations (Ma 157c); Morse theory and Floer homology (Ma 191b).

## Research Interests

Low-dimensional topology: knot theory, Khovanov homology, knot Floer homology, categorification; geometric representation theory: geometric Langlands.

## Research Experience

**PCMI Participant** Jun. 2026 – Jul. 2026
*Mentors: Megan Fairchild* IAS

Student in the Undergraduate Summer School (USS) at the 2026 IAS/Park City Mathematics Institute (PCMI). Explored knot Floer homology of $(1,1)$-knots. Proved that for $a = 1$ the $[p, a, b, r]$ Heegaard diagram represents a $(1,1)$-knot which both satisfies the Alexander criterion and is an $L$-space knot. Met and discussed with subject experts Ciprian Manolescu and Jacob Rasmussen.

**WWS Participant [S1]** Jan. 2026 – Jan. 2026
*Mentors: Dugan Hammock, Zsombor Méder* Wolfram

Student in the 2026 Wolfram Winter School (WWS). Explored S combinator arithmetic with a metamathematical basis using the Wolfram Language. Visually and symbolically defined the natural numbers using the $n$-tails of the 343-bird and introduced some dot notation for representing the combinator language.

**SURF Fellow** Jun. 2025 – Aug. 2025
*Mentors: Yi Ni, Daren Chen* Caltech

Student in the Geometry and Topology group at Caltech for the 2025 Summer Undergraduate Research Fellowship (SURF) program. Defining Bar-Natan homology for homologically essential links in $\mathbb{RP}^3$ and proving it is a link invariant. Exploring geometric implications such as defining a Rasmussen-type $s$-invariant and proving some slice genus bound. Will resume project in the spring of 2027.

**Undergraduate Researcher [P1]** Apr. 2025 – Jun. 2025, Apr. 2026 – Jun. 2026
*Mentors: Matthew Gherman* Caltech

Student in the Algebra group at Caltech for the spring of both 2025 and 2026. Introduced the theory of magic squares and, in particular, magic squares of squares. Used elementary number theory to exploit a series of magic square invariants to develop a single Diophantine equation whose solutions are in correspondence with the collection of radical magic diamonds. Reframed the existence/nonexistence of a magic square of squares through several equivalent statements.

**MSRI-UP Participant** Jun. 2024 – Jul. 2024
*Mentors: Candice Price, Erica Graham, Issa Susa* SLMath

Student in the 2024 Mathematical Sciences Research Institute Undergraduate Program (MSRI-UP) at the Simons Laufer Mathematical Sciences Institute (SLMath). Expanded on Margolskee et al.'s 2013 model of the menstrual cycle and Arbelaez-Gomez et al.'s 2022 model of endometrium growth to develop a proposed mathematical model of the cyclic relationship between estradiol and endometriosis lesions. Used MATLAB and Simulink to simulate endometrium and lesion volume growth as well as demonstrate the increased estradiol-to-progesterone ratio in a person with endometriosis. Analyzed the effectiveness of periodic exogenous estradiol suppression in preventing excessive endometrial tissue volume in the presence of lesions.

**WELP:U Participant [S2]** Sep. 2023 – Mar. 2024
*Mentors: Peter Barendse* Wolfram

Student in the 2023 Wolfram Emerging Leaders Program: Undergraduate (WELP:U). Explored the $3 \times 3$ magic squares of squares problem with an educational basis using the Wolfram Language. Used a series of magic square invariants to prove the fundamental form of magic squares of squares, constructed several unique radical magic hourglasses of squares, and geometrically interpreted the hypothetical existence of a magic square of squares.

**WSS Participant [S3]** Jun. 2023 – Jul. 2023
*Mentors: Brad Klee* Wolfram

Student in the New Kind of Science (NKS) and Ruliology track at the 2023 Wolfram Summer School (WSS). Explored methods of visualizing generalized Collatz functions with an experimental basis using the Wolfram Language. Observed numerous phenomena including structured distributions of prime number counts in iteration graphs, patterns in digit plots of different bases, and forced trajectories using truncated $p$-adic integers.

**SURF Fellow** Jun. 2022 – Aug. 2022, Apr. 2023 – Jun. 2023
*Mentors: Nathan Lewis, Sean Byrne* Caltech

Student in the Lewis Group at Caltech for the 2022 Summer Undergraduate Research Fellowship (SURF) program. Explored the effects of elemental substitution on cuprous oxide. Ran chronoamperometry tests with a PAR 273 potentiostat at varying potentials for pure and doped cuprous oxide solutions—specifically iodine and bromine inclusions. Then, characterized the deposited electrode films via X-ray diffractometry (XRD), and used the Wolfram Language to analyze the resulting data. Resumed the project for the spring of 2023, running chronopotentiometry tests with a tri-potentiostat system for cobalt inclusions and using scanning electron microscopy (SEM) for characterizations.

## Technical Experience

**IN FOCUS Participant** Jan 2024 – Jan 2024
*Mentors: N/A* Jane Street

Student in the Trading track at the 2024 IN FOCUS program. Learned quantitative trading concepts, developed market making language, and simulated market interactions through games like Figgie.

**MSP Fellow** Jul. 2023 – Sep. 2023
*Mentors: Regina Eckert* NASA JPL

Student in the Imaging Spectroscopy group (382B) at NASA JPL for the 2023 Maximizing Student Potential in STEM (MSP) program. Produced synthetic time-series VSWIR data for next-generation algorithm development by creating a prototype image segment selection tool, RT-Finder, using Python. Used the Skimage library for image processing by applying a series of Felzenszwalb segmentation, simple linear iterative clustering (SLIC) segmentation, and regional adjacency graph (RAG) thresholding to a given image. Then, used the OpenCV library to create an interactive window for segment selection, with the generated image masks being converted into a single, integer array to be inpainted with data from the SHIFT campaign read along EMIT wavelengths.

## Preprints and Publications

(1) Russell Martinez, *Generating Radical Magic Diamonds*, preprint, 2026, `https://russmanmath.github.io/Paper1.pdf`

## Selected Articles

(1) *Birdwatching: A tale of S combinator arithmetic*, Wolfram Community (2026), `https://community.wolfram.com/groups/-/m/t/3629162`

(2) *Exploring magic squares of squares*, Wolfram Community (2024), `https://community.wolfram.com/groups/-/m/t/3188586`

(3) *Exploring generalized Collatz functions*, Wolfram Community (2023), `https://community.wolfram.com/groups/-/m/t/2959684`

## Tutoring and Mentoring

**Mathematics Tutor** Jan. 2022 – June 2025
*Supervisors: Liz Jackman* Caltech Y

Mathematics tutor in the Caltech Y Rise Program. Demonstrated principles of math and their applications to K-12 students in local Pasadena, California. Advised one student in particular through all four years of high school, helping them achieve self-proficiency in pre-college and International Baccalaureate (IB) mathematics.

**Mathematics Tutor** Jan. 2022 – Mar. 2023
*Supervisors: Annice Jackson* Caltech Y

Mathematics tutor in the Caltech Y Young Legends Tutoring Program. Demonstrated principles of math and their applications to K-12 students associated with the San Gabriel Valley Section of the National Council of Negro Women (SGV-NCNW).

## Talks and Presentations

(1) *Knot Floer Homology of* $(1,1)$*-Knots*, presented at PCMI, IAS, Jul. 2026

(2) *Elementary Topos Theory and Intuitionistic Logic*, presented at Introduction to Discrete Mathematics (Ma 6c), Caltech, Jun. 2026

(3) *Birdwatching: A Tale of S Combinator Arithmetic*, presented at WWS, Wolfram, Jan. 2026

(4) *On Knot Floer Homology*, presented at Selected Topics in Mathematics: Morse theory and Floer Homology (Ma 191b), Caltech, Mar. 2024

(5) *The Bunkbed Conjecture: How to Debunk a Bunkbed*, presented at Combinatorial Analysis (Ma 121a), Caltech, Dec. 2024

(6) *Proposed Mathematical Model Of The Cyclic Relationship Between Estradiol And Endometriosis Lesions*, presented at MSRI-UP, SLMath, Jul. 2024

(7) *Creating Synthetic Time-Series VSWIR Data for Next Generation Algorithm Development*, presented at MSP, NASA JPL, Sep. 2023

(8) *Effects of Elemental Substitution on Cuprous Oxide Crystallinity*, presented at SURF Seminar Day, Caltech, Oct. 2022

## Awards

(1) Featured Contributor, awarded by Wolfram Community, Jan. 2026

(2) Stanley and Chenmei Hsu SURF Fellow, awarded by Caltech, Jul. 2025

(3) Featured Contributor, awarded by Wolfram Community, Jun. 2024

(4) Featured Contributor, awarded by Wolfram Community, Jun. 2023

(5) Dr. and Mrs. Daniel C. Harris SURF Fellow, awarded by Caltech, Jul. 2022